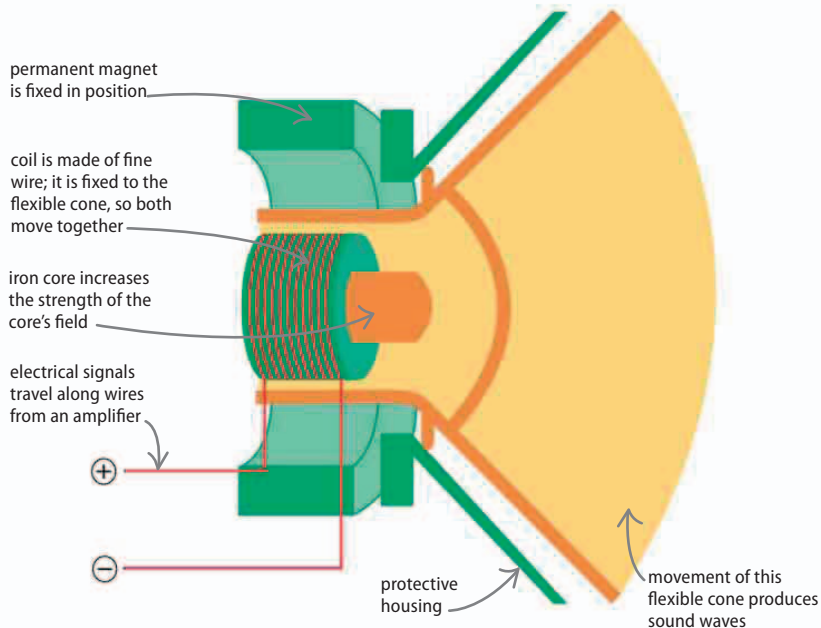


Loudspeaker

A loudspeaker uses the motion generated by the forces between a permanent magnet and an electromagnet to reproduce sound. Fluctuating electric current enters the coil, producing a fluctuating magnetic field. The forces between this field and that of the permanent magnet move the coil rapidly in and out. The coil moves the cone and these movements generate sound waves.



△ Electromagnetism in action

The forces acting on the moving parts of a loudspeaker are electromagnetic, produced by the interaction of the permanent magnet and the coil electromagnet.

REAL WORLD

Robotic arm

The joints and parts of an industrial robot arm, such as this car welding robot, are powered by electric stepper motors. A central rotor can be turned in steps by the magnets, making the motor capable of very precise movements.



The world's smallest electric motor is just 1nm (1 nanometer) across.

Linear motor

This type of electric motor creates a force in a straight line rather than the turning force of a traditional rotary motor. It achieves this by a continuous sequence of magnetic attraction and repulsion between electromagnets along a track and magnets attached to a sled, train, or some other object running along the track. The electromagnets repeatedly switch their polarity to move the object down the track without the need for wheels.

▷ Magnetic motion

Maglev (magnetic levitation) trains use powerful magnets to float above a track and are propelled forward at great speed by a linear motor.

